06 The Mutation Room

Mutation is random with respect to fitness

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 33 min	**+ 32 min**	The Selection Room (05), for what s means. The Fate Room (03) helps with step 1.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#mutation`

Four wheels, one per character of a fish, decide which derived state a mutation produces. Some slices are larger than others: some mutations are simply easier to make. The new alleles then enter a population of N as single copies. This guide separates the two questions the room keeps apart: which mutations appear, and what becomes of them.

LEARNING OBJECTIVES

After the core path you should be able to

- Distinguish the probability that a mutation *arises* from the probability that it *spreads*.
- Predict that new neutral alleles are usually lost, and that selection raises the chance of favoured ones.
- Show, from the wheels, that selection does not change which mutations are produced.
- Explain the phrase “mutation proposes, selection disposes” with evidence from the room.

The extension adds

- Use $u(1/N) = (1 - e^{-2s})/(1 - e^{-2Ns})$ to predict fixation bars, and judge them against sampling error.
- Explain why the ratio between selected and neutral fixation grows with N.
- Recognise mutation bias in the wheels' slice sizes and say why it is not the same thing as a directed mutation.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Lecture demonstration:** step 2, freezing the wheels on screen while the class watches the ★ slices.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Spin, and watch four new alleles

8 min

SET UP

Population size $N = 50$, *Fate of the alleles*: **Neutral**.

PREDICT FIRST

Each of the four derived alleles starts as one copy among 50. How many of the four will be fixed by the end of the run?

RUN

Press **Spin — mutate all four characters**, then watch the *Derived alleles* panel. Spin three more times.

LOOK AT

the four frequency lines, and the fish after G generations.

EXPLAIN

Was your prediction right? Connect the result to what you saw in the Fate Room.

2 Turn selection on

10 min

SET UP

Fate of the alleles: **Selection**. The stream favours a slender body, blue colour, a long tail and short fins (marked ★). Raise s to about 0.2.

PREDICT FIRST

Two things to predict separately. (a) Will the ★ slices of the wheels get bigger? (b) Will the favoured alleles do better in the population?

RUN

Spin several times. Watch the wheels while you raise and lower s, and watch the frequency lines after each spin.

LOOK AT

the size of the ★ slices, and which lines climb and which are pushed down.

EXPLAIN

Which of your two predictions came true? What does that tell you about where selection acts?

3 A hundred runs of the same four mutations

10 min

SET UP

Selection on, $s = 0.1$, $N = 50$. Spin once.

PREDICT FIRST

For the four alleles you just drew, predict the height of each bar in *Fixed in 100 runs*. Which will be near the dashed neutral line at $1/N$, and which above or below it?

RUN

Press **Run 100 Simulations**. Then switch back to Neutral and press it again.

LOOK AT

each bar against the dashed line at $1/N = 0.02$.

EXPLAIN

Why did one run (step 2) tell you so much less than a hundred?

4 Put it in one sentence

5 min

SET UP

Nothing to change.

PREDICT FIRST

Before writing, look back at your answers to steps 2 and 3.

EXPLAIN

Write one sentence that uses the words *mutation*, *selection*, *origin* and *fate*, and that someone who has not seen this room would understand.

EXTENSION · MSC

5 Predicting the bars

12 min

SET UP

$N = 50$, Selection on. Use $s = 0.05$, 0.1 and 0.2 in turn, with the same spin.

PREDICT FIRST

Calculate $u(1/N)$ for a favoured allele at each s. How big is the sampling error of a bar from 100 runs?

RUN

Run 100 Simulations at each s and read the bars for the favoured alleles.

EXPLAIN

Do the bars agree with your calculation within sampling error? Why is $u \approx 2s$ a reasonable shortcut here?

6 Population size changes the contrast

10 min

SET UP

s = 0.1, Selection on. N = 50, then N = **300**.

PREDICT FIRST

What will happen to the neutral line at $1/N$? To the bars of favoured alleles? To unfavoured ones?

RUN

Run 100 Simulations at each N.

EXPLAIN

The ratio of selected to neutral fixation is about $2Ns$. Why does a larger population sort mutations more sharply, even though it receives the same four mutations?

7 Mutation bias

10 min

SET UP

Neutral. Reset between spins if you like.

PREDICT FIRST

Pick one wheel, for example body colour. If you spin twenty times, how often will each derived state come up? Use the slice sizes.

RUN

Spin twenty times and tally the outcomes for that wheel.

EXPLAIN

Some states are produced more often than others. Is that a bias towards *useful* states? How could a strong mutation bias still influence which traits evolve, even though selection does not set it?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc "Bacteria mutate to become resistant when they meet an antibiotic." What is wrong with this sentence, and how would you rewrite it?
2. BSc Why is the observation that a favoured allele usually wins *after* it appears not evidence that the environment *caused* it to appear?
3. BSc The room assumes one mutation per character. Real mutation rates are around 10^{-8} per site per generation. Which of your conclusions would change with realistic rates, and which would not?
4. MSc Explain the Luria–Delbrück fluctuation test, and say what it showed about the relationship between mutation and selection.
5. MSc The room's wheels have unequal slices. Give a real example of mutation bias (for example transitions against transversions, or CpG mutability) and discuss whether it can shape evolution.
6. MSc The neutral theory says most substitutions are neutral even though most adaptive change is selected. How do the bars in step 6 help reconcile those two statements?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** Most spins end with all four lines on the floor. Each new neutral allele fixes with probability 1/50, so on average 0.08 of the four fix per spin.
- **Step 2.** The ★ slices never change size, whatever s is. Favoured alleles climb more often and the rest are pushed down: selection acts after the mutation exists.
- **Step 3.** Neutral bars sit near 0.02. With $s = 0.1$, favoured bars rise to about 0.18 and unfavoured ones fall towards zero.
- **Step 4.** A good answer: “Mutation decides which variants appear, blindly; selection only changes the fate of a variant once it has appeared.”
- **Step 5.** $u \approx 0.096$, 0.18 and 0.33 for $s = 0.05$, 0.1 and 0.2. A bar from 100 runs has a standard error of $\sqrt{(u(1-u)/100)}$, about 0.03–0.05 here.
- **Step 6.** At $N = 300$ the neutral line falls to about 0.003, favoured bars stay near 0.18, and unfavoured ones vanish. The contrast between selected and neutral grows with N.
- **Step 7.** The tallies follow the slice sizes. The bias is in what is *easy to make*, not in what is useful; but a state produced often is offered to selection more often, so bias can matter when several routes would serve.